MILITARY SPECIFICATION SHEET

ELECTRON TUBE, KLYSTRON
TYPE 8634

The complete requirements for procuring the product described herein shall consist of this document and the issue in effect of Specification MIL-E-1.

DESCRIPTION: CW amplifier, 1.0 kW, 4.4 - 5.0 GHz, 4 integral cavities, individual tuners, waveguide input and output, forced-air cooling, permanent-magnet focused.

OUTLINE AND DIMENSIONS: See Figure 1 herein. WEIGHT: 52 lbs., approx.

ABSOLUTE RATINGS: (Note 1)

Parameter:	Ef	Eb	Load	tk	If (surge)	Pi	Iby
Unit:	V	kVdc	VSWR	min.	a	kW	mAdc
Maximum:	6.8	8	2:1	—	14	3.8	60
Minimum:	6.0	—	—	3	—	—	—

Parameter:	Ambient Oper. Pressure	Ambient Temp.		Collector Temp.	Oper. Temp.		
		Oper.	Stg.		Collector	Body	Tuner Fins
Unit:	atmosphere	°C	°C	°C	°C	°C	°C
Maximum:	—	+55	+85	+325	+200	+150	+150
Minimum:	0.67	-31	-65	—	—		
				(Note 2)	(Note 2)		

TEST CONDITIONS: (Note 3)	Ef	Eb	Load	tk	Oper. Temp. (Notes 2, 3)		
					Collector	Body	Tuner Fins
	V	kVdc	VSWR	min.			
	6.5	7.5	1.05	3	+150to +200°C	+125to+150°C	+125to+150°C

BASE: See Figure 1 herein.

RF INPUT CONN.: Mates with UG-149A/U flange, or equal.

RF OUTPUT CONN.: Mates with UG-149A/U flange, or equal

MOUNTING POSITION: (Preferred-) Vertical, cathode down.

COOLING: Forced air. (Note 3)

FOCUSING: Permanent magnet. (Note 4)

HANDLING CAUTION: (Note 5)

FSC-5960

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T15:08Z
Check the source to verify that this is the current version before use.

MIL-E-1 Method or Ref.Par.	Requirement or Test	Conditions (See Appx E-50.1)	AQL (% def.)	Insp. Level or Code	Symbol	Limits Min.	Limits Max.	Unit
	GENERAL							
—	Qualification	Not required; (Note 6)	—	—	—	—	—	—
—	First Article Testing	Required; (Note 7)	—	—	—	—	—	—
3.1.1	Performance	(Note 8)	—	—	—	—	—	—
3006	Cooling	(Note 9)	—	—	—	—	—	—
E-50.2	Holding period	t=168 hrs.	—	—	—	—	—	—
D-20	Dimensions	(Fig. 1)	—	—	—	—	—	—
3.7	Marking	(Note 10; Fig.1)						
	QUALITY CONFORMANCE INSPECTION, PART 1, (NOTE 11)							
3.6 D-20	Visual - Mechanical examination	—	—	—	—	—	—	—
—	Body current	Power Output (2)	0.65	II	Iby	—	30	mAdc
1256	Cathode current	—	0.65	II	Ik	425	500	mAdc
1301	Heater current	—	0.65	II	If	7.2	8.2	A
4211	Power Output(1):	F=4.4, 4.7, 5.0 GHz Pd=10 mW	0.65	II	Po Po/Pi	1.0 28	— —	kW %
—	Gain				Po/Pd	50	—	db
4211	Power Output(2):	F=4.4, 4.7, 5.0 GHz Pd=40 mW (Note 12)	0.65	II	Po Po/Pi	1.2 32	— —	kW %

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T15:08Z
Check the source to verify that this is the current version before use.

MIL-E-1 Method or Ref.Par.	Requirement or Test	Conditions (See Appx E-50.1)	AQL (% def.)	Insp. Level or Code	Symbol	Limits Min.	Limits Max.	Unit
	QUALITY CONFORMANCE INSPECTION, PART 2							
D-20	Dimensions	(See Fig.1)	—	—	—	—	—	—
4201	Bandwidth	F=4.7 GHz Pd=50 mW Po=1 kW,min Freq.Deviation= ± 3.5 MHz (Note 13)	2.5	L6				
—	Tuner operation:	T_A= + 25°C	6.5	L6				
	Mech. tuning range	(Note 14)			—	—	—	—
	Stop torque	(Note 15)			—	—	—	—
	Starting torque	(Note 16)			—	—	—	—
1031	Sweep-frequency vibration	No voltages applied (Notes 17,18)	—	—	—	—	—	—
—	Shock	No voltages applied (Notes 18,19)	—	—	—	—	—	—
1016	Moisture resistance	(Note 20)	—	—	—	—	—	—
1021	Temperature cycling	Test Cond.A. except T(low)= -65°C (Note 18)	—	—	—	—	—	—
	QUALITY CONFORMANCE INSPECTION, PART 3, LIFE, (Note 21)							
4.7; 4.7.4	Life test	Group D Power Output(2)	—	—	t	1000	—	hrs
4.7.3	Life-test end-point:				Po	0.96	—	kW
5.1	Preparation for delivery	(Note 22)	—	—	—	—	—	—
1136	Rough-handling test	Per MIL-E-75 or contract	—	—	—	—	—	—

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T15:08Z
Check the source to verify that this is the current version before use.

NOTES:

1. Refer to Specification MIL-E-1, Section 6 "Note" covering absolute maximum ratings.

2. Collector temperature measured at (center of collector) bottom of fin, in collector top, nearest beam axis. Under "no drive" condition, maximum temperature is effective from maximum collector dissipation. At any ambient temperature, care must be taken to keep the maximum collector temperature below + 200°C when tube is delivering 1 kW. (See Figure 1 herein.)

3. Relative to the tube-temperature requirements specified herein, the following cooling-means accomplishment is recommended:

Air Flow			Inlet Air Pressure	
Collector (ducted) cfm	Body (free) cfm	Tuner (ducted) cfm	Collector in.-H_2O	Tuner in.-H_2O
200	60	60	2.0	0.25

4. All connectors should be of non-magnetic material. In general, care shall be exercised to prevent demagnetization: ferromagnetic material shall not be brought within 6 inches of the tube, and any other magnets shall not be brought within 12 inches of the tube.

5. Care should be exercised during all handling for installation, storage, and use of the tube, to prevent impartation of shock or vibration in excess of the values for which the tube is designed.

6. The Preproduction Sample Approval requirements in U. S. Army Electronics Command Drawing SC-A-46600B hereby replace any Qualification requirements referable to the product covered herein. However, the requirements of paragraphs 3.1.2.1, 3.1.2.2, 3.1.3, and 4.4 in Drawing SC-A-46600B shall be considered not applicable to the product covered herein. (See Note 10, below.)

   (Copies of U. S. Army Electronics Command Drawing SC-A-46600B, 19 March 1965, "Preproduction Sample Approval In Lieu Of Qualification Requirements in Specifications For Semiconductor Devices And Electron Tubes" may be obtained from the procuring activity or as directed by the contracting officer.)"

7. The term "First Article Testing" shall be considered synonomous with the term "Preproduction Sample Approval" as used in Note 6 above. All requirements and tests pertinent to Preproduction Sample Approval as covered in Drawing SC-A-46600B shall be considered as identically applicable to First Article Testing (and associated requirements) established herein.

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T15:08Z
Check the source to verify that this is the current version before use.

8. Associable with and in addition to the performance requirements covered herein, all tests listed in Specification MIL-E-1, paragraph 3.1.1, are applicable except those per (therein-) referred paragraphs 3.6.2, 3.6.3, E-50.3.1, E-1201, E-1211, D-40, D-20b(2), E-1131, E-1216 and 3.5.2, E-1101, E-1111, E-1006, and E-40.1(a).

9. The cooling requirements specified or otherwise covered herein shall be effective wherever applicable throughout the testing program herein.

10. Marking shall be in accordance with Specification MIL-E-1. If any specification-requirements waiver has been granted, the product-identification marking shall consist of the tube type-number only.

11. The AQL for the combined defectives for attributes in Quality Conformance Inspection, Part 1, exclusive of Visual-Mechanical examination, shall be 1.0%.

12. With respect to sequential testing from Power Output (1) test to performance of Power Output (2) test, the following procedure shall apply: With the tube synchronously tuned (respectively, at the designated frequency), cavity #3 shall be set approximately 1/4-turn higher in frequency; drive power shall then be increased to that specified and then cavity #3 shall be retuned to maximize power output; the other cavities may be then adjusted to maximize power output.

13. Standard "stagger" tuning procedure shall be employed as follows: cavities #1 and #4 shall be tuned at designated frequency; cavity #2 shall be set below, and cavity #3 shall be set above, the designated frequency.

14. The tuner of each cavity shall require approximately 12 complete turns to cover the frequency range 4.4 to 5.0 GHz.

15. The tuning mechanism of each cavity shall be capable of withstanding, at the extreme ends of the tuning range, a stop torque of 200 oz/in.

16. The starting torque for the tuner of each cavity shall be less than 30 oz/in. This test shall be performed immediately following the Stop Torque test per Note 15 herein.

17. The tube shall be mounted rigidly on the vibration table with the waveguide supported by struts. Vibration shall be simple harmonic motion. The tube shall be vibrated through a complete scan cycle in each of three mutually perpendicular planes, one of which shall be parallel to the beam axis. Each scan cycle shall be of ten minutes duration and consist of a continually varying vibration frequency between the limits of 10 and 500 cps. The vibration frequency shall be divided into three (3) steps with peak accelerations respectively as follows:

G = 10; between 10 and 100 cps
G = 5; between 100 and 150 cps
G = 2.0; between 150 and 500 cps

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T15:08Z
Check the source to verify that this is the current version before use.

18. At the completion of this test, the tubes shall meet all of the electrical test requirements of Quality Conformance Inspection, Part I, herein.

19. The tube shall be subjected to a total of nine (9) impact shocks, three (3) shocks in each of three mutually perpendicular planes, X, Y, and Z. The Z plane is defined as the vertical plane with heater connector pointing down. Each shock pulse in the Z plane shall have an acceleration of 20 G for a duration of 11 $\pm$ 1 milliseconds with a peak acceleration not to exceed 30 G. Each shock pulse in the X and Y planes shall have a peak acceleration of 20 G with a duration of 11 $\pm$ 1 milliseconds measured at the 1/4-amplitude of the shock pulse.

20. Four hours after completion of this test, the tube shall meet all of the electrical test requirements of Quality Conformance Inspection, Part 1, herein.

21. Compliance with applicable Life-test sampling and testing will be met on the following basis:

a. At any one time for a particular lot, a minimum of one sample tube shall be under Life test.

b. For successive lot(s), a minimum of one sample tube shall be subjected to Life test and this shall be done immediately following completion of Life test performed on the preceding sample tube(s). The sample tube(s) so subjected shall, where the Life-test interval coincides, be selected from the latest or current lot presented for inspection acceptance.

c. Each sample tube subjected to Life test shall be identified with respect to the lot(s) which it represents.

22. Unless otherwise specified in contract or order, each tube shall be sealed within a moisture-vaporproof barrier-material bag normally acceptable under packaging requirements of Specification MIL-E-75.

23. Tubes subjected to the following destructive tests shall not be delivered on contract or order:

Life Test

Custodian:

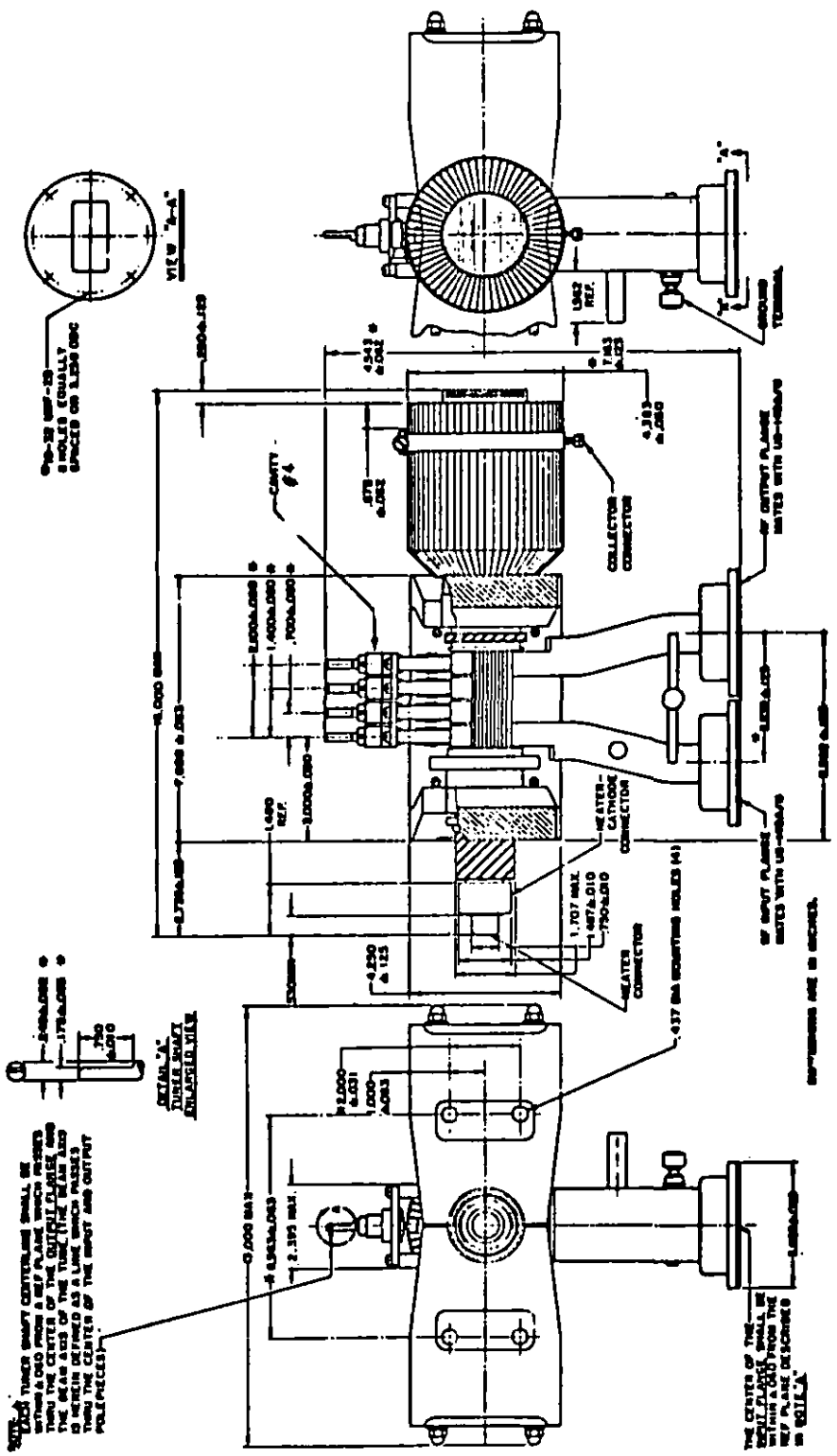
Army-EL

Preparing activity:

Army-EL

Project No. 5960-A627

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T15:08Z
Check the source to verify that this is the current version before use.



NOTES: 1. All dimensions in inches. Tolerances on decimal dimensions are ± .005 unless otherwise specified.
2. Inspection requirements for dimensions marked with an asterisk=0.65% AQL, Insp. Level II; all other dimensions=6.5% AQL, Insp. Level L6.
3. Tube shall be furnished with protective dust covers on input and output flange.
4. Tube serial number shall be marked in location readily readable when tube is mounted.

Figure 1. Outline and dimensions.

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T15:08Z
Check the source to verify that this is the current version before use.

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

DD FORM 1 OCT 64 1426

ESC-FM 2011-66

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T15:08Z
Check the source to verify that this is the current version before use.